

3 (short) Figure 4 shows a radio frequency tuner circuit connected to an antenna. The antenna has an internal resistance of $50\ \Omega$. The tuner circuit consists of a $15.5\ \mu\text{H}$ inductor and a variable capacitor, C .

(a) The variable capacitor is adjusted to $3.1\ \text{nF}$. If the antenna produces an r.m.s. voltage v_{in} of $100\ \text{mV}$ at $513\ \text{kHz}$, find the magnitude of the current through the capacitor and its phase with respect to the voltage v_{in} at the antenna. [6]

(b) Determine the capacitance of the variable capacitor required to tune the circuit to receive BBC Radio Cambridge at $1.026\ \text{MHz}$. [4]

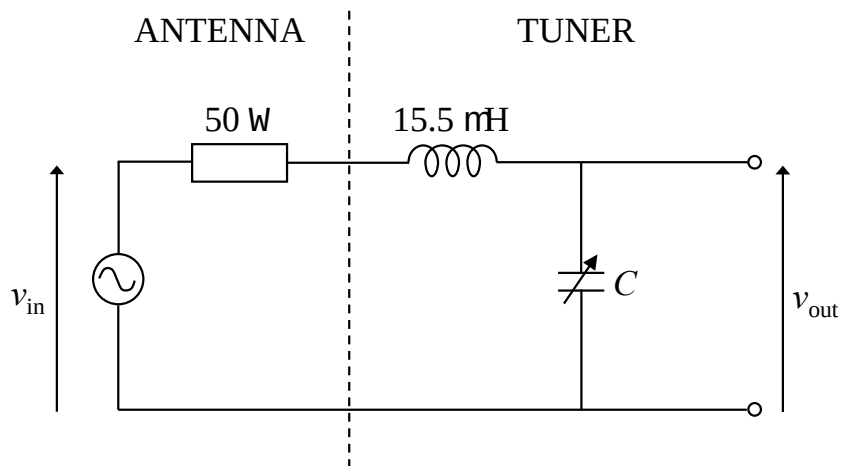


Fig. 4